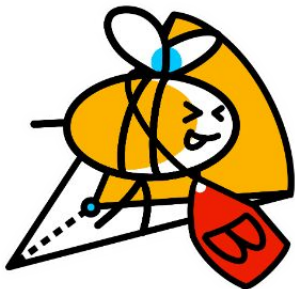
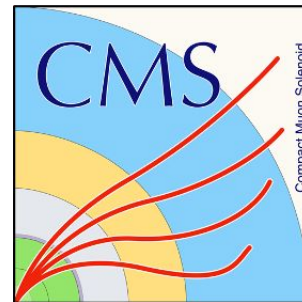


CMS-DAS BTV exercise

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HATS 2025
LPC, FNAL
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Introduction



- BTV stands for B-Tagging and Vertexing
 - “This group is responsible for all aspects of b tagging, including algorithm development, performance measurement with data, HLT and the corresponding online and offline DQM tasks. It is also responsible for vertex fitting and secondary vertex finding.”
- Wiki page: <https://btv-wiki.docs.cern.ch/>
- Subgroups
 - Software & Algorithms
 - Performance, Validation, Commissioning
 - HLT (High-Level Trigger)
- Closely working with JME (Jets and Missing Energy) POG



Flavour tagging

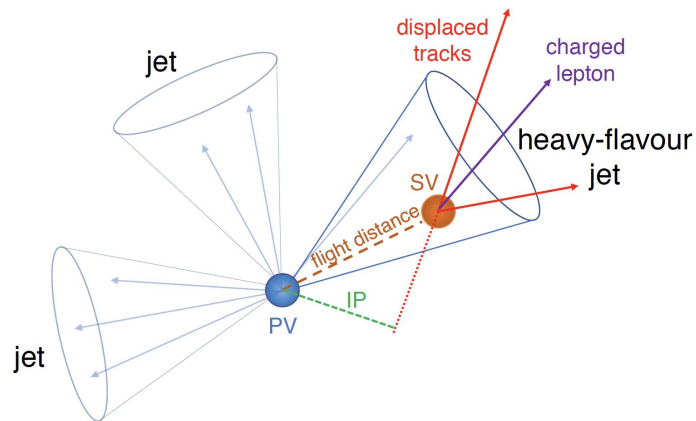
- In CMS, particles are reconstructed using the ParticleFlow (PF) algorithm
- Quarks and gluons result in jets (clusters of particles)
 - Reconstructed with the Anti- k_T algorithm from individual PF particles
 - Single-pronged tagging with AK4 jets (quarks, gluons)
 - Multi-pronged tagging with AK8 jets (multiple quarks, e.g. coming from a boosted Z/W/t/Higgs and collimated)



Flavour tagging



- b and c quarks have special detector signatures
- Useful for many analyses (Higgs, top quarks)
 - Need to reject large-crosssection events (e.g. QCD) composed mostly of light quarks (u, d, s) or gluons
- B hadrons (from b) and c hadrons (from c, mainly D hadrons) have large lifetime, large mass, travel a bit through the detector before decaying
- Displaced decay: Displaced tracks and secondary vertices
- In ~20% of b hadron and ~10% of c hadron decays, we find a lepton (muon/electron)

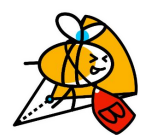




Use Machine Learning!



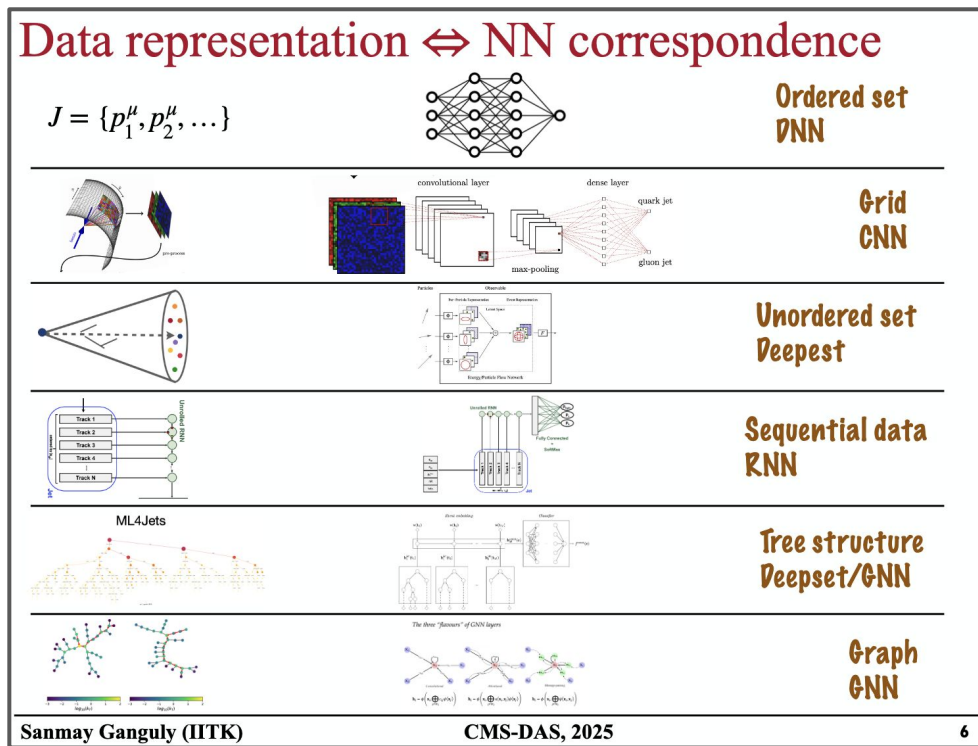
- Jets contain several particles
(charged or neutral)
- Each particle has several properties
(impact parameter, mass, charge...)
- Try to combine all this information to
predict whether the jet came from a
b, c, or other quark/gluon



Use Machine Learning!



- Jets contain several particles
(charged or neutral)
- Each particle has several properties
(impact parameter, mass, charge...)
- Try to combine all this information to
predict whether the jet came from a
b, c, or other quark/gluon
- Try to combine as much information
as possible!



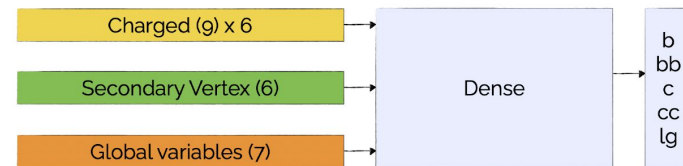


ML algorithms (Run-2)



- DeepCSV:

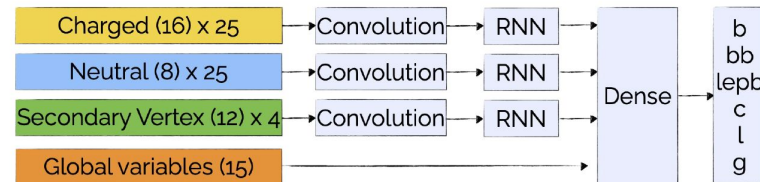
- Combine information of upto 6 tracks (a.k.a. charged particles)
- Simple deep NN/MLP (4 layers) architecture
- Around 66 inputs



- DeepJet:

- Process charged, neutral and SV particles separately with CNN and RNNs
- More particles and more properties per particle
- Combine all information in the end with an MLP
- Around 600 inputs

DeepJet





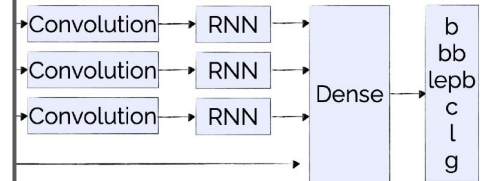
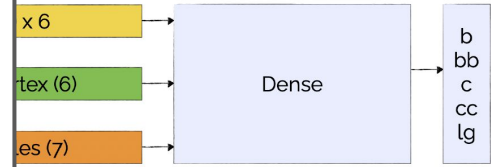
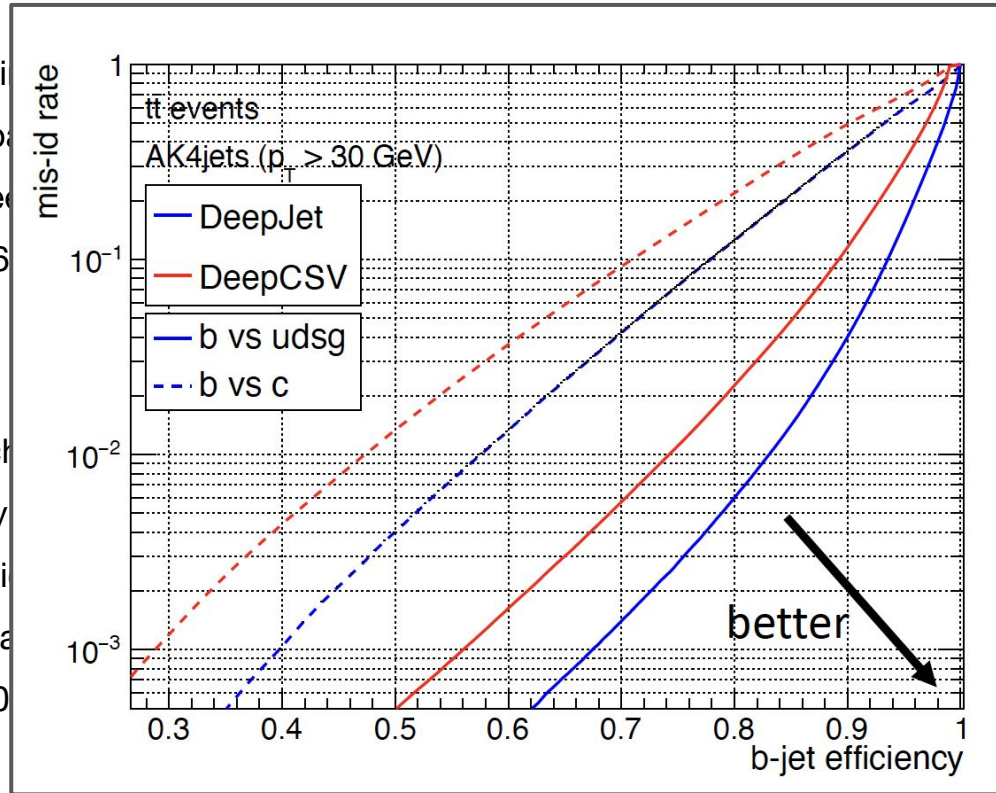
ML algorithms (Run-2)

- DeepCSV:

- Combine information from charged particles
- Simple deep neural network
- Around 66% efficiency

- DeepJet:

- Process charged particles separately
- More particles
- Combine information
- Around 60% efficiency

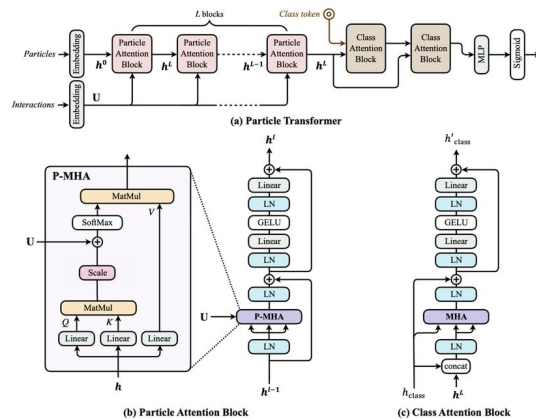
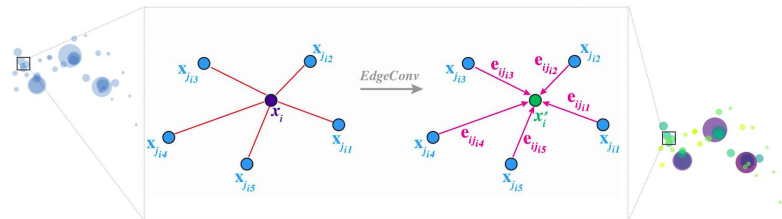




ML algorithms (early Run-3)



- ParticleNet:
 - Treat jet as an unordered graph of particles
 - Find latent abstract representations iteratively using graph convolutions: Dynamic Graph Convolutional Neural Networks (DGCNNs)
- ParticleTransformer:
 - Use MultiHeadAttention and particle pairwise interactions: Transformer architectures
 - Use adversarial training to improve robustness against MC mismodelings

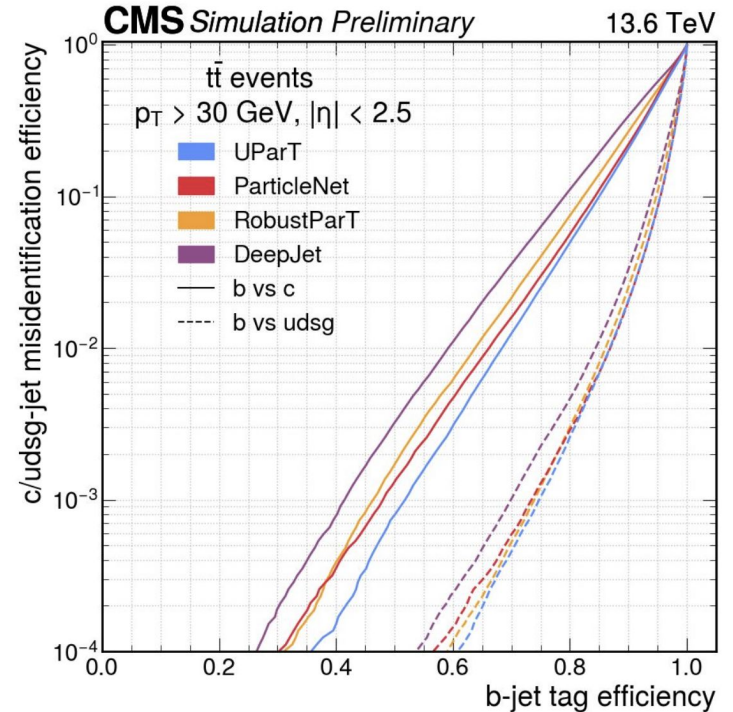




ML algorithms (2024)



- Unified Particle Transformer:
 - Larger model
 - Larger training dataset
 - Additional targets, including hadronic taus
 - Adversarial training
 - Simultaneous prediction of jet energy and resolution as regression targets
 - State of the art starting 2024 data reprocessing (NanoAOD v15)





ML algorithms



- All algorithms are trained as multiclassifiers:
 - Same model with a c probability node for c-tagging
- Similar algorithms for AK8 jets, e.g. ParticleNetAK8, GlobalParticleTransformerAK8



Exercise 1



- Important branches in NanoAOD (for MC)
 - Jet_hadronFlavour
 - Jet_partonFlavour

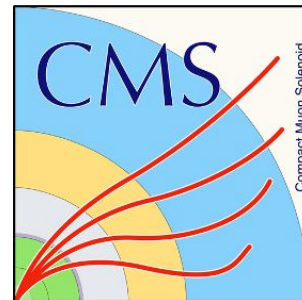
Jet flavor	label	Definition	
b	bb	hadronFlavor=5	$N(\text{b-hadrons}) > 1$
	b		$N(\text{b-hadrons}) = 1$ && not $(\Delta R(\text{nu frp, b/c hadron, jet}) < 0.4)$
	lepb		$N(\text{b-hadrons}) = 1$ && $\Delta R(\text{nu frp, b/c hadron, jet}) < 0.4$
c	c	hadronFlavor=4	
uds	uds	hadronFlavor!=5 && hadronFlavor != 4	partonFlavor=1 or 2 or 3
g	g		partonFlavor=21



Exercise 1



- “True” flavours and tagger “scores” (outputs):
 - Tagger scores: can be interpreted as probability for a jet to be of a certain class
- Discriminants (redefine scores based on what we want):
 - $$\text{discriminant } X \text{ vs } Y = \frac{p(X)}{p(X) + p(Y)}$$
 - Stored in NanoAOD. Exercise helps you access and plot them.
- Working point: Cut on the discriminant score to select a pure sample of X (with minor Y contribution)



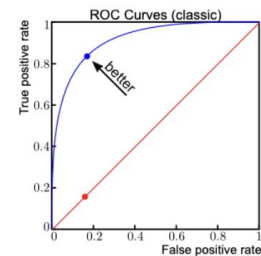
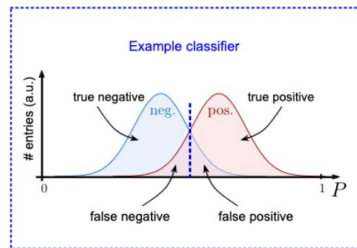
Your turn!



Exercise 2

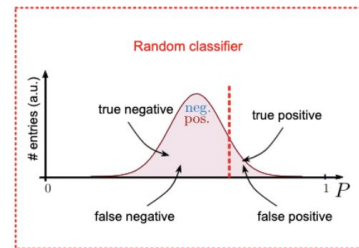
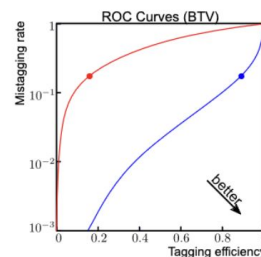


- How good is our tagger?
 - We usually look at receiver operating characteristic (ROC) curves and the area under curve (AUC) value

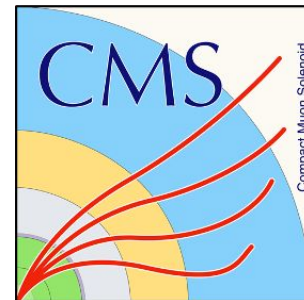


AUC: area under curve
(e.g. under the ones on the left
←)

— $0.5 < \text{AUC} < 1$
— $\text{AUC} = 0.5$



A. Stein



Your turn!